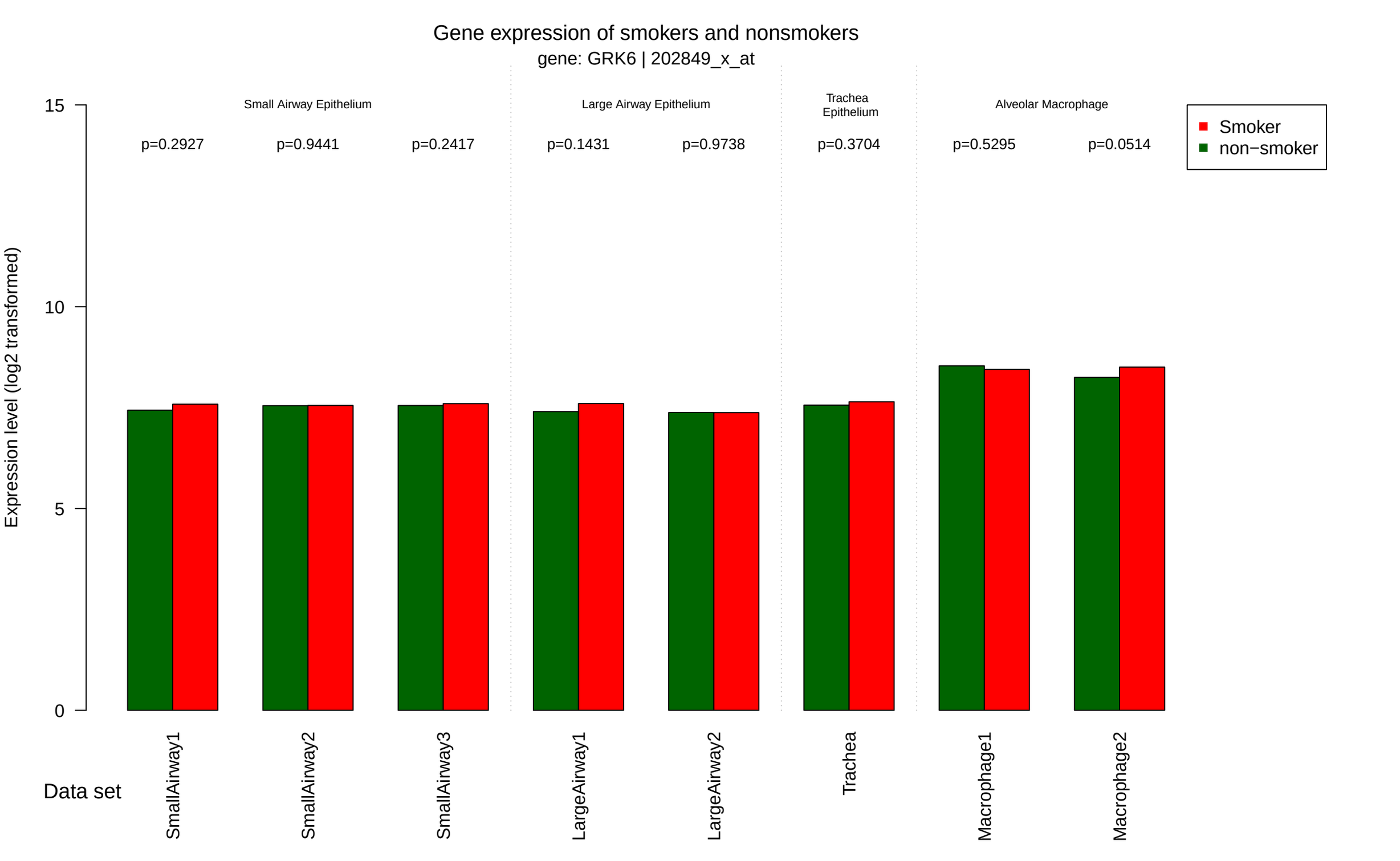
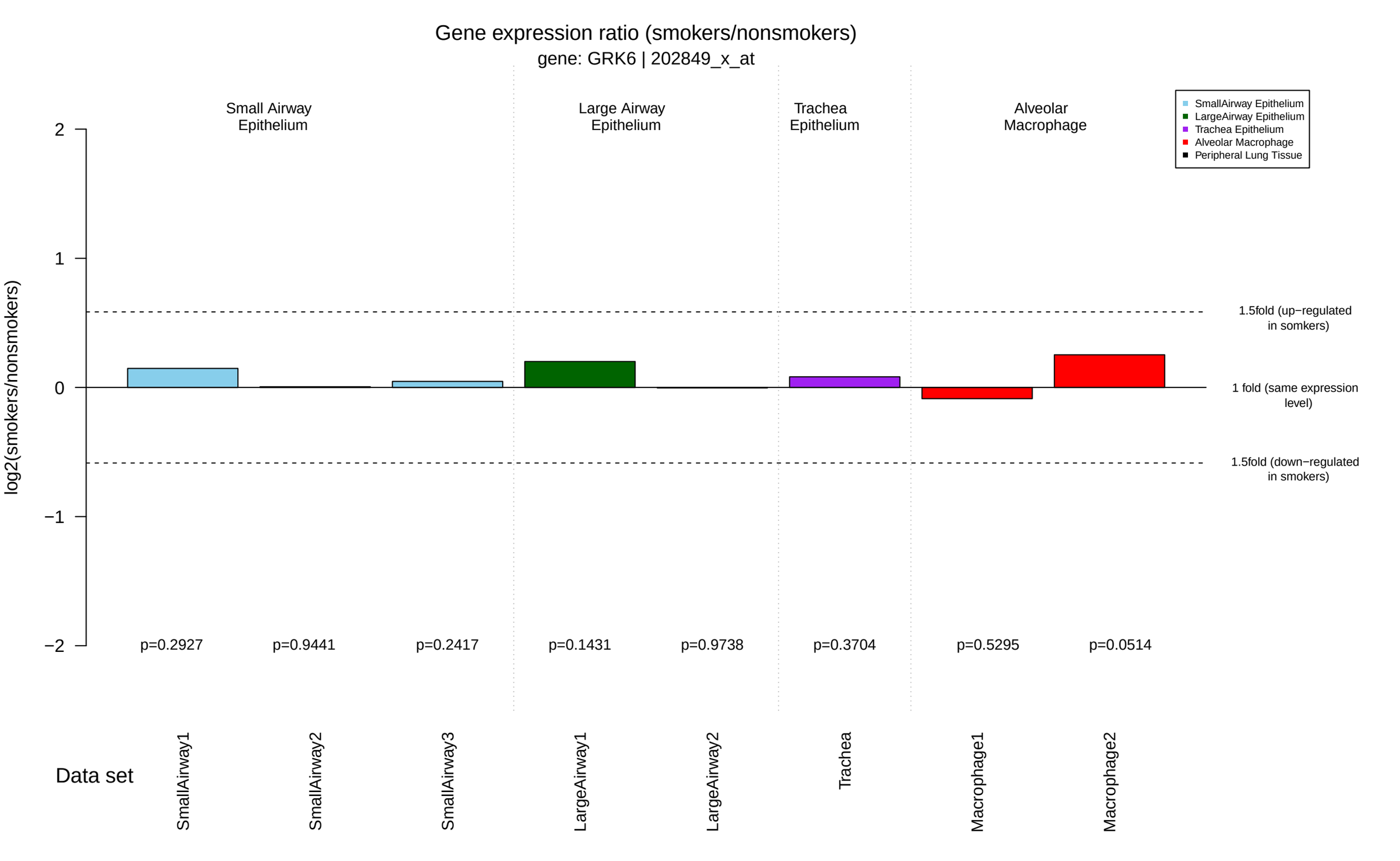
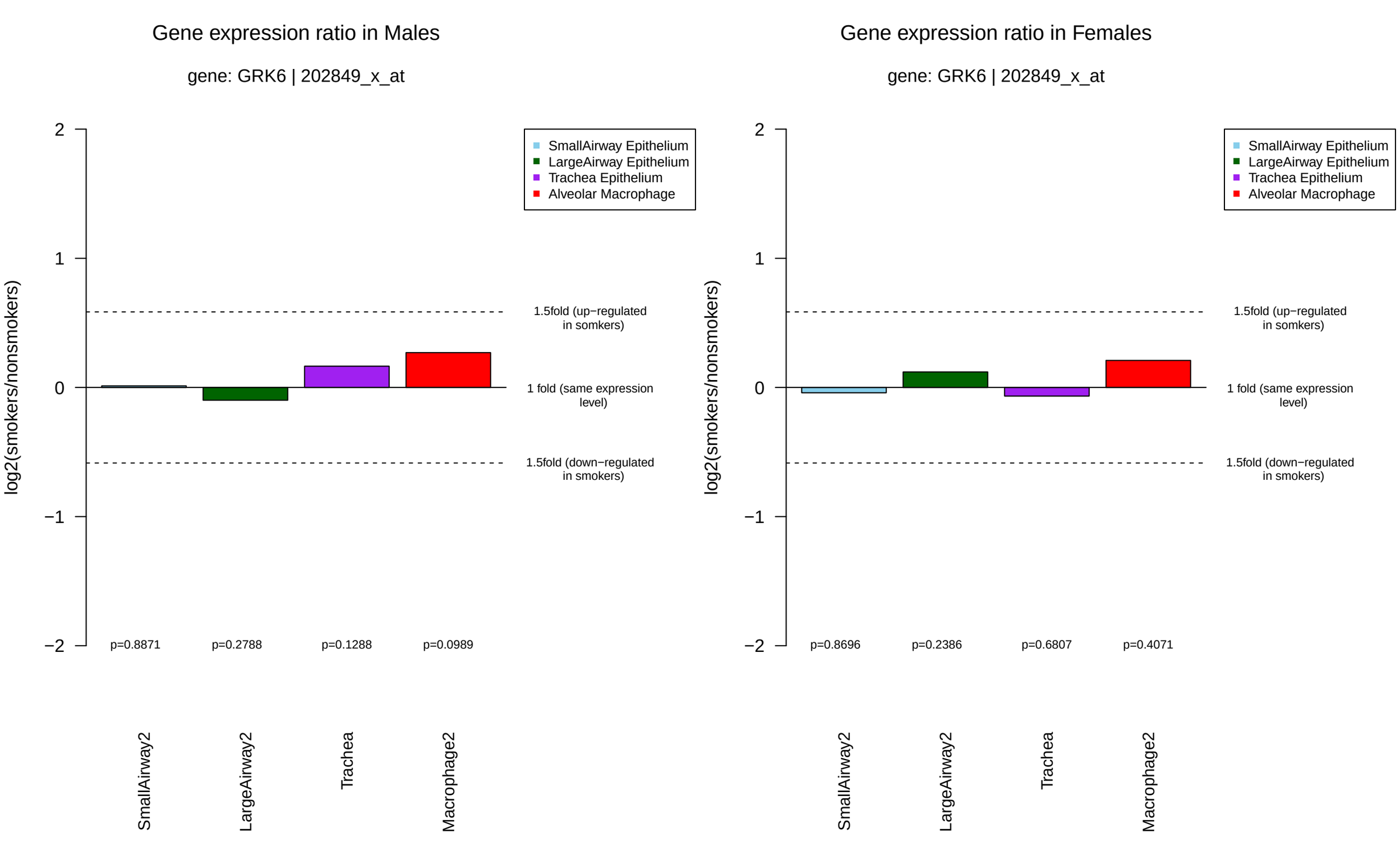
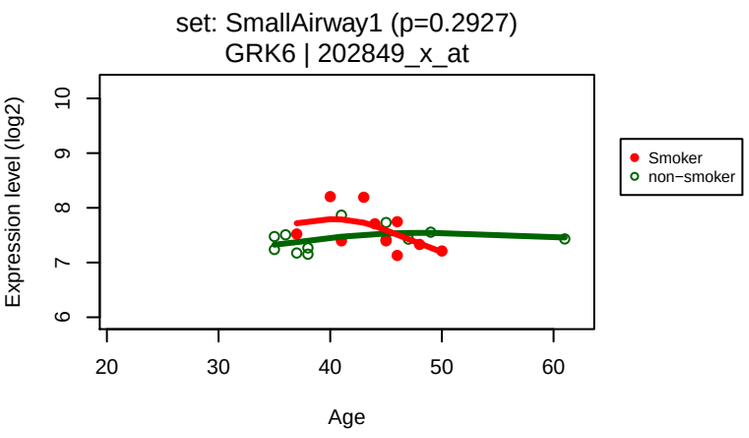
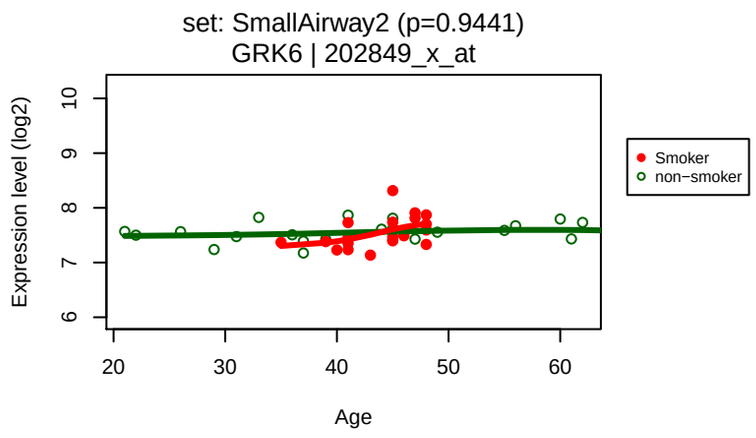
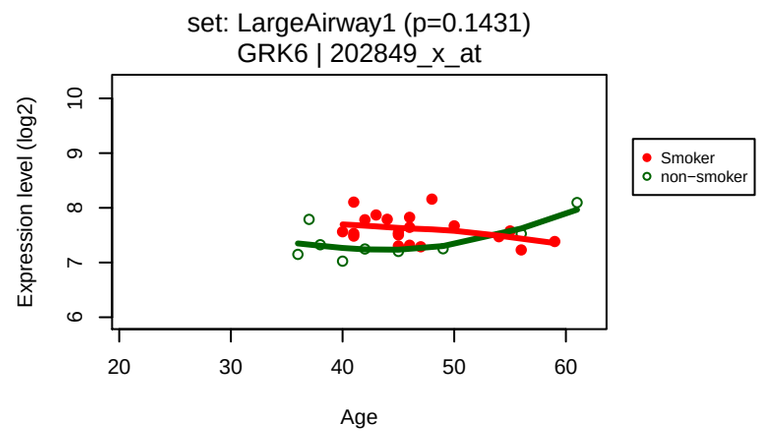

Gene expression of smokers and nonsmokers
gene: GRK6 | 202849_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2927
p=0.9441
p=0.2417
p=0.1431
p=0.9738
p=0.3704
p=0.5295
p=0.0514
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

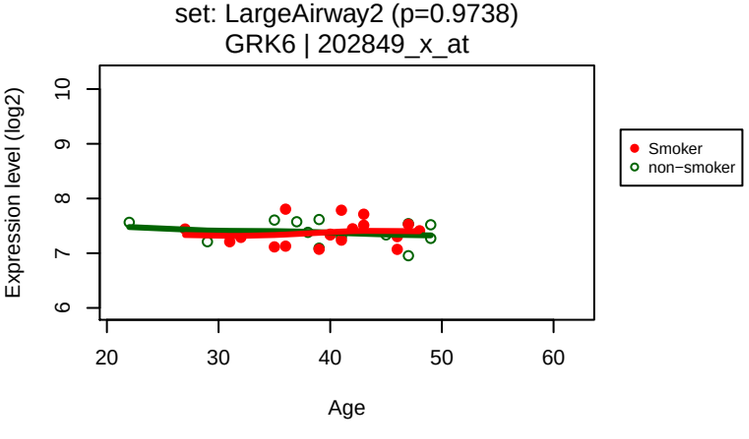

Gene expression ratio (smokers/nonsmokers)
gene: GRK6 | 202849_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2927
p=0.9441
p=0.2417
p=0.1431
p=0.9738
p=0.3704
p=0.5295
p=0.0514
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

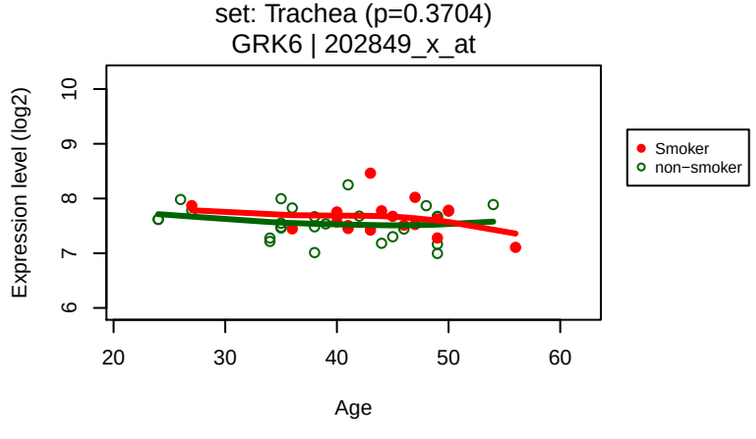

Gene expression ratio in Males
gene: GRK6 | 202849_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8871
p=0.2788
p=0.1288
p=0.0989
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRK6 | 202849_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8696
p=0.2386
p=0.6807
p=0.4071
SmallAirway2
LargeAirway2
Trachea
Macrophage2

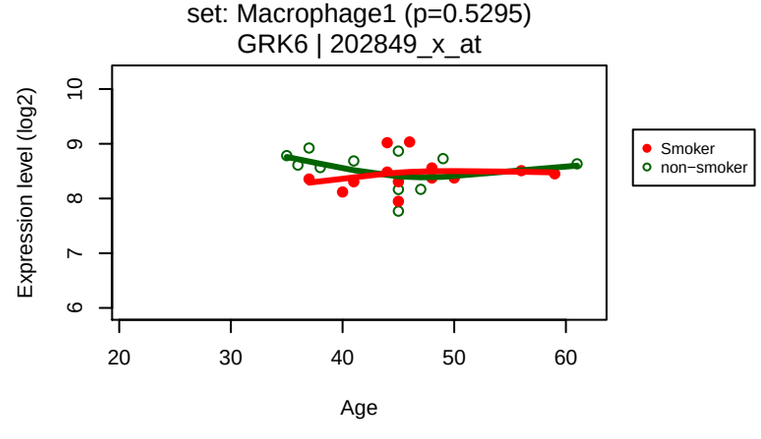

set: SmallAirway1 (p=0.2927)
GRK6 | 202849_x_at
Expression level (log2)
Age
Smoker
non−smoker

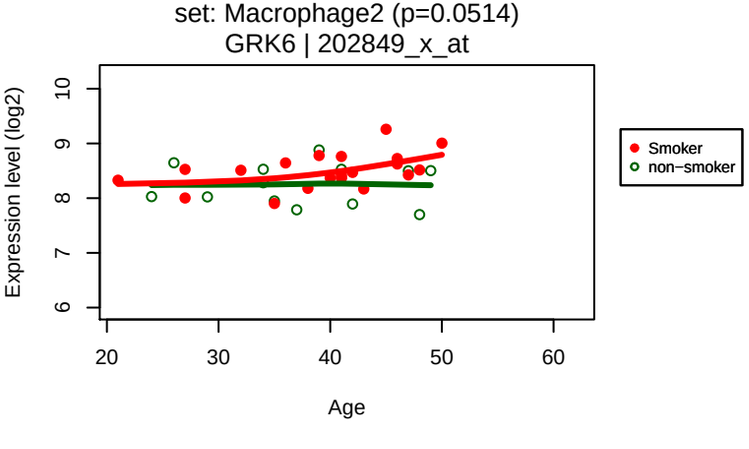

set: SmallAirway2 (p=0.9441)
GRK6 | 202849_x_at
Expression level (log2)
Age
Smoker
non−smoker

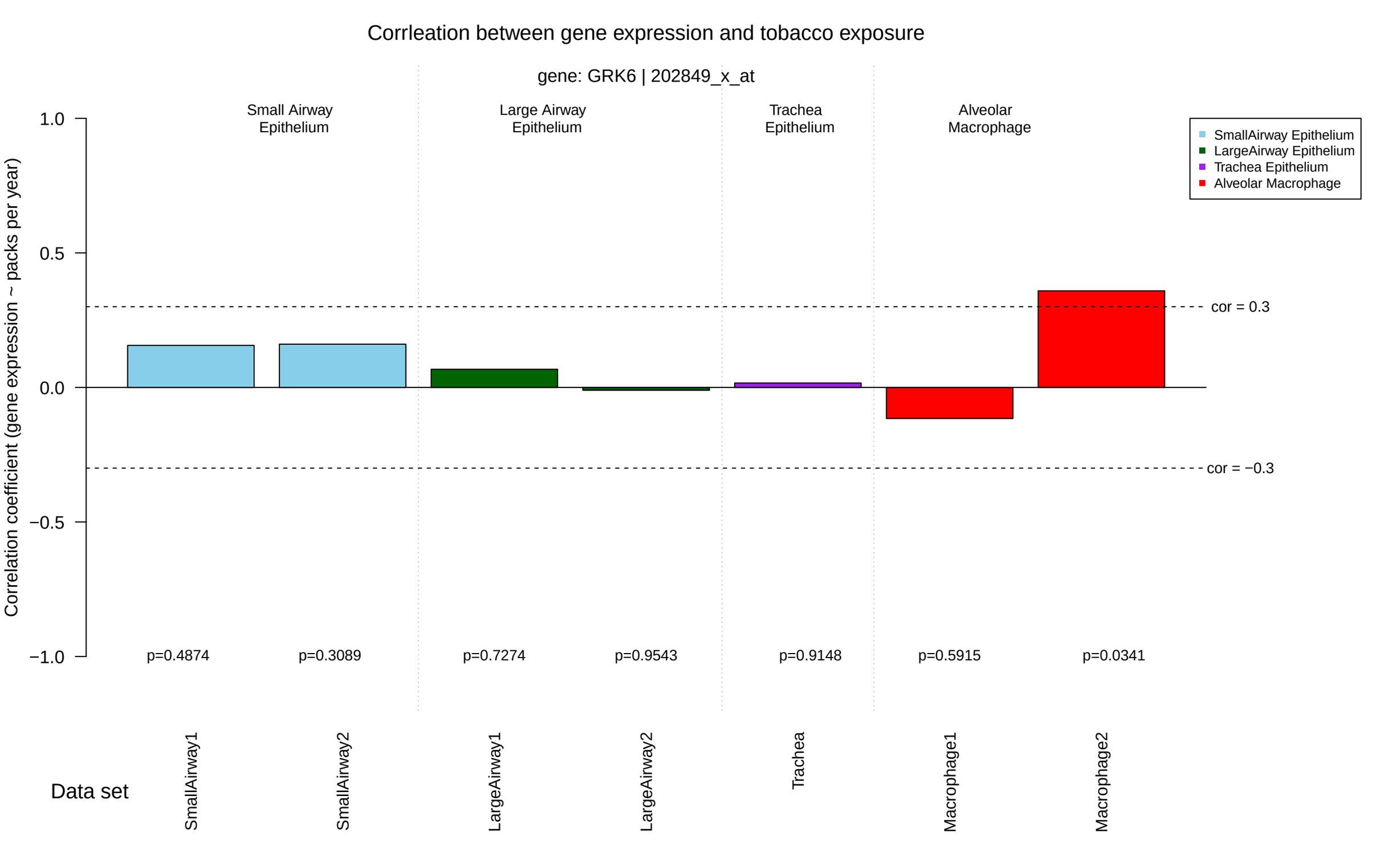

set: LargeAirway1 (p=0.1431)
GRK6 | 202849_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9738)
GRK6 | 202849_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3704)
GRK6 | 202849_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5295)
GRK6 | 202849_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0514)
GRK6 | 202849_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRK6 | 202849_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4874
p=0.3089
p=0.7274
p=0.9543
p=0.9148
p=0.5915
p=0.0341
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2